PROJECT TECHNICAL MANUAL

Louisiana Land Degradation Explorer

Interactive Google Earth Engine Application for Parish-Level Vegetation Decline Screening (2001–2025)

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Implementation basis: Louisiana_Land_Degradation_Explorer.js, checked against the supplied Earth Engine Code Editor screenshot.

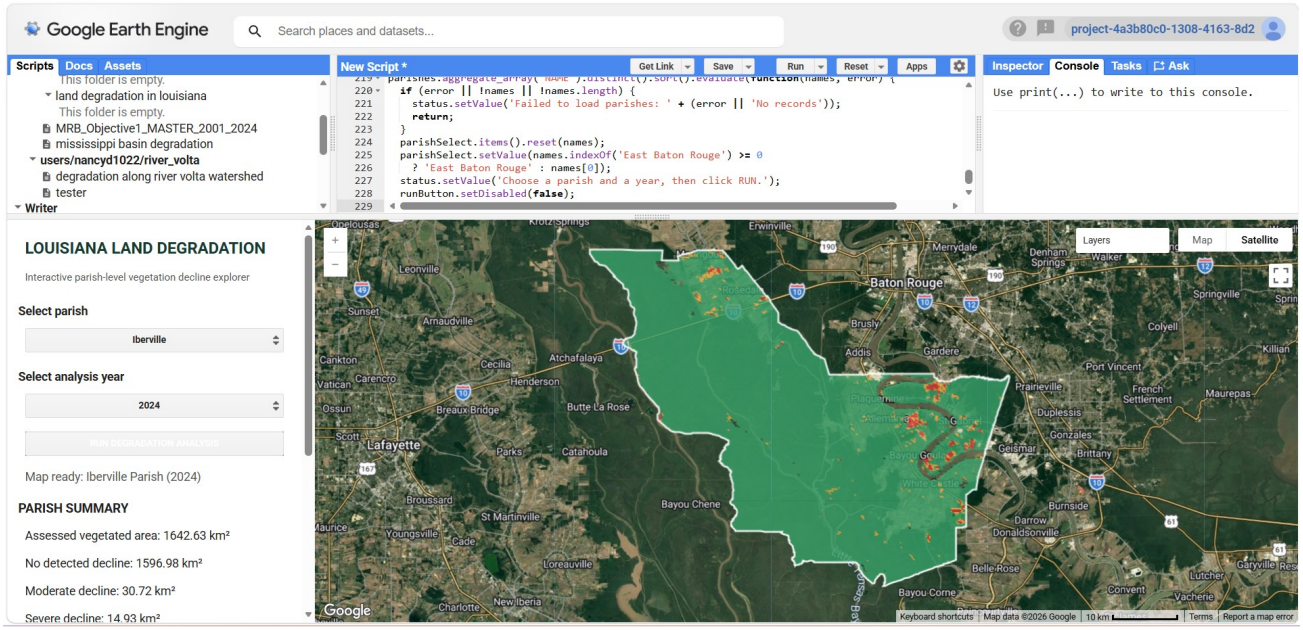


Figure 1. Working application in the Earth Engine Code Editor, showing Iberville Parish for 2024 (user-provided screenshot).

Executive summary

The Louisiana Land Degradation Explorer is a Google Earth Engine (GEE) application that screens potential declines in vegetation condition at Louisiana parish level. A user chooses a parish and calendar year, then the application quality-filters MODIS Terra vegetation indices, constructs annual median NDVI and EVI images, compares each against a fixed 2001–2005 baseline, and maps three mutually exclusive classes. The application reports areas in square kilometres and exports a classified GeoTIFF.

The mapped classes are indicators of vegetation decline—not conclusive diagnosis of land degradation. Land conversion, floods, drought, timber harvest and marsh dynamics can produce similar signals. The technical specifications below describe the existing script exactly, including its assumptions and limitations.

1. Objectives, scope and intended applications

- Locate potential moderate and severe negative changes in vegetation condition relative to a fixed historical reference.
- Allow interactive selection of each of Louisiana's 64 parishes and a year from 2001 through 2025.
- Summarise mapped class areas, degraded-share indicator, and export a georeferenced raster for subsequent GIS evaluation.
- Support preliminary screening, field reconnaissance planning, environmental reporting, and hypothesis generation; not regulatory determination of degradation.

2. System architecture and workflow

GEE datasets → MODIS quality masking and scale-factor correction → annual median NDVI/EVI → five-year baseline → index declines → baseline vegetation eligibility and valid-data mask → dual-index thresholds → three-class map → parish area reduction → interface and export.

3. Study area and spatial definition

The study covers the State of Louisiana, USA. The administrative polygons are provided by the 2018 U.S. Census TIGER county-equivalent dataset. In Louisiana, the county equivalents are parishes. The script retains features where STATEFP = "22" and obtains parish names from the NAME field. A selected parish polygon supplies the analysis geometry for clipping, rendering, regional reduction and GeoTIFF export.

Although a parish selection covers the whole parish polygon, results are limited to the subset eligible under the vegetation baseline and current-year valid-data conditions; total parish area is not used as the percentage denominator.

4. Datasets and variables

Dataset / asset	Identifier and details	Role in code
MODIS Terra Vegetation Indices v6.1	MODIS/061/MOD13Q1; nominal 250 m; 16-day composites; years analysed 2001–2025	Source of NDVI, EVI and DetailedQA bands
U.S. Census TIGER Counties 2018	TIGER/2018/Counties; polygon FeatureCollection; STATEFP,	Louisiana filter, 64-parish dropdown, clip and export region

	NAME	
GEE derived pixel area	ee.Image.pixelArea(); square metres converted to km²	Area sums for each class
Google satellite/hybrid background	ui.Map().setOptions("HYBRID")	Cartographic contextual basemap only; not an analytical input

4.1 MODIS product characteristics

MOD13Q1 Collection 6.1 is a Terra sensor-derived 16-day vegetation-index product. Each input image contains scaled integer NDVI/EVI values; the script multiplies both by 0.0001 to obtain dimensionless physical index values. The nominal pixel dimension is 250 m, corresponding to approximately 0.0625 km² per fully covered 250 m × 250 m square; actual area totals are computed from per-pixel area rather than assuming constant surface area.

The application uses only bands NDVI, EVI and DetailedQA for computations. Available reflectance bands are not loaded into the analysis. MODIS supplies computed indices; the program does not recalculate indices from reflectance.

4.2 Quality control

The MODLAND VI quality flag occupies bits 0–1 of the DetailedQA band. The code computes bitwiseAnd(3).eq(0), accepting only QA code 0 ("VI produced with good quality"). Values 1, 2 and 3 are masked out. This filter is deliberately strict, and retained valid observations can differ across years and landscapes.

$$Q_{i,t} = 1 \text{ if } (\text{DetailedQA}_{i,t} \text{ AND } 3) = 0; \text{ otherwise } 0$$

The code does not explicitly enforce minimum counts of observations, apply separate land/water QA bits, or harmonise MODIS with other sensors. Those are enhancement opportunities rather than current operations.

5. Mathematical basis: vegetation indices

5.1 Normalized Difference Vegetation Index (NDVI)

$$NDVI = (\rho NIR - \rho Red) / (\rho NIR + \rho Red)$$

ρNIR and ρRed denote near-infrared and red surface reflectance. NDVI typically ranges approximately from −1 to +1. Higher positive values generally indicate greater photosynthetically active green vegetation, whereas low or negative values can represent sparse vegetation, built surfaces or water. In this app, NDVI is read directly from MOD13Q1 and multiplied by 0.0001.

5.2 Enhanced Vegetation Index (EVI)

$$EVI = 2.5 \times (\rho NIR - \rho Red) / (\rho NIR + 6\rho Red - 7.5\rho Blue + 1)$$

ρBlue is blue-band surface reflectance. The gain G = 2.5, aerosol coefficients C_1 = 6 and C_2 = 7.5, and canopy background adjustment L = 1 are the conventional MODIS three-band EVI coefficients. MODIS may also use a backup two-band EVI calculation for certain problematic high-reflectance conditions, following its

production algorithm. The application consumes the provider-supplied EVI product and does not choose between these formulations itself.

6. Temporal compositing and baseline design

6.1 Annual median composites

For a selected integer year y, the script filters the quality-masked MODIS collection from 1 January (inclusive) to 1 January of the following year (exclusive). For each index, GEE calculates the pixelwise median of the retained 16-day composites. This produces a single NDVI and EVI image per calendar year.

$$VI_y(p) = \text{median}\ \{VI_t(p) : t \in \text{calendar year } y,\ Q_t(p) = 1\}$$

p is a pixel, VI is NDVI or EVI, and t represents a composite time. Missing or invalid observations are masked. Median aggregation limits the influence of individual anomalous composites but does not equate to a growing-season-only index or remove all phenological effects.

6.2 Five-year fixed reference

The baseline comprises annual medians for 2001, 2002, 2003, 2004 and 2005. GEE then takes the pixelwise median of those five annual image values, separately for NDVI and EVI. This is a median-of-annual-medians method, rather than a single median over all 16-day scenes in the five-year period.

$$VIref(p) = \text{median}\ \{VI_{2001}(p), VI_{2002}(p), \ldots, VI_{2005}(p)\}$$

A 2001–2005 analysis-year selection is compared with a baseline that contains that same year, making early-year comparisons partially self-referential. Therefore, early-year losses are not statistically independent of the baseline and need additional care in interpretation.

7. Change detection and classified degradation indicators

7.1 Absolute vegetation-index decline

$$\Delta NDVI_y(p) = NDVIref(p) - NDVI_y(p)$$

$$\Delta EVI_y(p) = EVIref(p) - EVI_y(p)$$

Positive Δ values mean the selected year has a lower vegetation index than the 2001–2005 reference. Negative Δ values indicate an increase relative to reference. These are absolute index-point changes, not percentages or trends.

7.2 Eligibility and valid-data mask

The code evaluates pixels only if baseline NDVI ≥ 0.25, baseline EVI ≥ 0.15, and both current-year index bands are unmasked. The baseline bands are implicitly required to be valid by their comparisons. This rule approximates “vegetated at reference,” and can include or exclude some land-cover conditions inappropriately, particularly in coastal marshes and wetlands.

$$V_y(p) = 1[NDVIref \geq 0.25 \wedge EVIref \geq 0.15 \wedge valid(NDVI_y) \wedge valid(EVI_y)]$$

7.3 Moderate and severe decision rules

Moderate and severe classes require the NDVI and EVI decline thresholds simultaneously. Severe takes precedence over moderate because the script assigns class 1 first and overwrites matching pixels with class 2.

$$M_\gamma(p) = 1[\Delta NDVI_\gamma \geq 0.08 \wedge \Delta EVI_\gamma \geq 0.05]$$

$$S_\gamma(p) = 1[\Delta NDVI_\gamma \geq 0.16 \wedge \Delta EVI_\gamma \geq 0.10]$$

$$C_\gamma(p) = 2 \text{ if } V_\gamma \wedge S_\gamma;\ 1 \text{ if } V_\gamma \wedge M_\gamma \wedge \neg S_\gamma;\ 0 \text{ if } V_\gamma \wedge \neg M_\gamma$$

A pixel that falls short on either index is assigned class 0, including pixels with decline in only one index. Thus class 0 means "no detected simultaneous decline under these thresholds," not "proven stable land." The threshold values are configurable constants in the script and were not supplied with independent calibration or field-validation evidence.

Code	Class	Map colour	Interpretation
0	No detected decline	#38a169 (green)	Eligible pixel below combined moderate thresholds
1	Moderate decline	#f6ad42 (orange)	Both moderate thresholds satisfied; severe condition not satisfied
2	Severe decline	#cc3030 (red)	Both severe thresholds satisfied
255	NoData (export only)	Not shown / uncoloured	Masked or outside evaluation area after unmask(255)

8. Spatial analysis and area estimation

For each class c ∈ {0,1,2}, the script masks the area image to pixels classified as c. GEE pixelArea() supplies area in m^2; division by 10^6 converts it to km^2. A sum reduction runs over the selected parish at nominal scale = 250 m, maxPixels = 10^{10} and tileScale = 4.

$$A_c = \Sigma_{p \in \text{parish}}\, 1[C_\gamma(p) = c] \times a(p) / 1{,}000{,}000 \ \ km^2$$

$$A_{assessed} = A_0 + A_1 + A_2$$

$$A_{potential} = A_1 + A_2$$

$$\text{Percentpotential} = 100 \times (A_1 + A_2) / (A_0 + A_1 + A_2)$$

Pixel area a(p) accounts for the analysis projection and edge pixels as evaluated by Earth Engine. Aassessed is not parish area; it includes only valid and eligible pixels. The percentage is therefore the proportion of assessed vegetation with detected joint index decline, not the proportion of the parish's total land area.

9. Illustrated application results: Iberville Parish, 2024

The supplied screenshot shows the interface reporting Iberville Parish for 2024. The visible statistics are transcribed below; a sample calculation using rounded displayed values is provided. The screenshot is a captured result, not an independently rerun verification.

Displayed summary item	Reported result
Assessed vegetated area	1,642.63 km^2
No detected decline	1,596.98 km^2

Moderate decline	30.72 km²
Severe decline	14.93 km²
Moderate + severe	45.65 km²
Share of assessed vegetation	≈ 2.78%

$$45.65\ \text{km}^2 = 30.72 + 14.93$$

$$2.78\% \approx 100 \times 45.65 / 1{,}642.63$$

The display values sum to 1,642.63 km² and agree to the shown decimal precision. Any minor discrepancy from full-resolution floating-point statistics may reflect rounding. The map shows green for no detected decline, orange for moderate and red for severe, on a satellite backdrop with a white parish outline.

10. Using the application: step-by-step

1. Open Google Earth Engine Code Editor at https://code.earthengine.google.com/ and authenticate with an Earth Engine-enabled Google Cloud project.
2. Create a new JavaScript script and paste the complete Louisiana_Land_Degradation_Explorer.js code. Save the script under a descriptive project name.
3. Click Run. Earth Engine loads the 2018 TIGER parishes and fills the Select parish dropdown from unique sorted NAME values.
4. Choose one Louisiana parish; the default is East Baton Rouge if present.
5. Choose a calendar year from 2001 through 2025; the default displayed selection is 2024.
6. Click RUN DEGRADATION ANALYSIS. The button is temporarily disabled during computation; the status label reports progress.
7. Inspect the coloured class overlay and white parish border. The map centres on the chosen parish.
8. Scroll down the panel to PARISH SUMMARY for assessed area, each class area, potential-decline area and percentage.
9. Use the map's Layers menu to enable annual NDVI or baseline-minus-year NDVI difference; these layers are initially hidden.
10. Click EXPORT CLASSIFICATION TO GOOGLE DRIVE; open the Tasks tab and click RUN on the newly created task. Choose export settings in the task dialog and allow the task to finish.
11. Download the GeoTIFF from Google Drive, and open it in ArcGIS Pro or QGIS. Interpret codes 0, 1, 2 and 255 with the legend above.

11. Interface, visualization and exports

The custom interface is a horizontal panel-plus-map layout. Parish and year are ui.Select controls; the run and export actions are ui.Button controls. A status label displays selected parish/year and processing outcome. Statistics are computed with reduceRegion and asynchronously delivered to the user interface via evaluate().

Default visible layer: potential-degradation class image, opacity 0.80; boundary rendered in white with a two-pixel line. Optional layer 1: annual NDVI, displayed across 0–0.85. Optional layer 2: NDVI decline, displayed across −0.2–0.25. The HYBRID map is a background service only.

Export.toDrive settings: image = class raster with NoData filled as integer 255 and converted to unsigned byte; region = selected parish geometry; scale = 250 m; fileFormat = GeoTIFF; folder = GEE_Louisiana_Degradation; description and prefix = LA_Degradation_<Parish_with_underscores>_<Year>.

No explicit CRS is set, so output CRS follows Earth Engine export behavior for the image. The task needs manual activation in the Tasks tab.

12. Reproducibility and configuration

Parameter	Value	Where used
START_YEAR / END_YEAR	2001 / 2025	Year dropdown
BASE_START / BASE_END	2001 / 2005	Reference image collection
SCALE	250 m	reduceRegion and export
MODERATE_NDVI / MODERATE_EVI	0.08 / 0.05	Moderate combined condition
SEVERE_NDVI / SEVERE_EVI	0.16 / 0.10	Severe combined condition
Baseline vegetation filter	NDVI ≥ 0.25; EVI ≥ 0.15	Valid analysis mask
MODIS quality code	DetailedQA bits 0–1 = 0	Input masking

For reproducible reporting, record the code version, analysis parish, year, date of processing, data catalog product version, thresholds, baseline years, QA rule, scale and all exported outputs. Earth Engine catalog availability may change; this documented script sets an end year of 2025 rather than dynamically selecting the latest complete year.

13. Interpretation, assumptions and limitations

- Land degradation is multidimensional. This model captures simultaneous negative change in vegetation indices, not direct measurement of soil erosion, salinity, soil organic carbon, wetland loss or declining agricultural yields.
- Annual median VI responses can arise from short-lived drought, flooding, hurricanes, land management or harvesting; causal attribution requires external information.
- The approach does not implement an explicit temporal persistence test, trend significance test, trajectory model or machine-learning classifier.
- The 250 m MODIS scale may blend built surfaces, cropland, forest, marsh and open water into mixed pixels; fine-scale local changes can be missed.
- The fixed 2001–2005 baseline includes the target year for 2001–2005 analyses; these baseline-period comparisons should not be regarded as independent change detection.
- The two-index AND rule is conservative with respect to single-index declines, but its sensitivity and specificity are unknown without validation.
- Parish-level eligible vegetation area can differ between years because valid current-year data masks differ; comparing percentages over time needs common-pixel masking or adjusted methods.
- Water or wetland surfaces and coastal land-water changes can alter the VI signal; a separate land/water mask or dedicated wetland-change layer would strengthen coastal inference.
- The displayed percentages are of the eligible assessed pixel area, not the full administrative polygon.
- 2018 parish geometries are used for every year, meaning possible administrative boundary modifications are not represented historically.

14. Quality assurance and recommended validation

12. Check that all 64 parish names load, and compare selected parish boundaries with U.S. Census mapping.
13. For one parish/year, calculate an independent pixel sample and inspect raw QA, scaled VI, baseline VI, loss and final class.
14. Verify 0, 1, 2 classes are mutually exclusive, and masked pixels become 255 in the export.
15. Compare sum of class areas with total assessed valid-mask area at consistent projection and scale.
16. Vary threshold pairs and the baseline interval to produce a sensitivity table of potentially degraded km² and percentages.
17. Compare maps with NLCD land-cover transitions, Landsat/Sentinel high-resolution composites, hurricane/flood footprints, drought indicators and field/georeferenced observations.
18. Develop a stratified reference sample of stable/moderate/severe or changed/unchanged sites; produce an error matrix, user's/producer's accuracies, and uncertainty intervals.
19. Use persistent multiyear decline and seasonally matched composites as alternative specifications to separate transient stress from degradation.

15. Suggested next-generation research upgrades

A robust research extension could add: (i) seasonally consistent Landsat/Sentinel VI metrics; (ii) rainfall-normalised productivity and drought anomalies; (iii) land-cover transitions and vegetation persistence; (iv) soil exposure, erosion and soil organic carbon proxies; (v) wetland/open-water dynamics, especially in coastal parishes; and (vi) an independently validated composite degradation score. These are proposed future layers and are not implemented in the documented script.

16. Troubleshooting

Observed issue	Likely cause / action
Parish dropdown remains loading	Check GEE project access, catalog availability, browser console and TIGER collection path.
Statistics fail or return no values	Check valid MODIS observations, selected year, collection availability, regional reduce limits and masks.
Map appears mostly green	Review legend and threshold conditions; green includes eligible pixels without joint decline, not assured healthy vegetation.
Some locations appear uncoloured	Baseline non-vegetated filter, masked observations or outside selected geometry.
Export not in Drive	Click RUN in GEE Tasks; confirm job success and Google Drive destination.
Different results when rerun	Record code edits, product availability, processing date, projection/scale choices and any threshold changes.

17. Data provenance and references

Google Earth Engine Data Catalog. MOD13Q1.061 Terra Vegetation Indices 16-Day Global 250m. https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1

Google Earth Engine Data Catalog. TIGER: US Census Counties 2018. https://developers.google.com/earth-engine/datasets/catalog/TIGER_2018_Counties

NASA LP DAAC / USGS. MODIS Vegetation Index User's Guide, Collection 6.1. https://lpdaac.usgs.gov/documents/621/MOD13_User_Guide_V61.pdf

Google Earth Engine documentation. Image.pixelArea, ImageCollection.median, Image.reduceRegion, Export.image.toDrive. https://developers.google.com/earth-engine/apidocs/

Implementation source: Louisiana_Land_Degradation_Explorer.js, supplied in the current project conversation. Screenshot: Google Earth Engine result for Iberville Parish, year 2024, supplied by the user.

Appendix A. Minimal pseudocode of the existing classification

20. Filter TIGER/2018/Counties to STATEFP == "22".
21. Read MODIS/061/MOD13Q1; keep DetailedQA bits 0–1 == 0.
22. Scale NDVI and EVI by 0.0001; calculate each selected year's per-pixel median.
23. Build baseline as median of annual 2001–2005 medians.
24. Compute loss as baseline minus current-year index values.
25. Keep pixels with baseline NDVI ≥ 0.25 and EVI ≥ 0.15 and valid current indices.
26. Class 2 when NDVI loss ≥ 0.16 AND EVI loss ≥ 0.10; else class 1 when NDVI loss ≥ 0.08 AND EVI loss ≥ 0.05; otherwise class 0.
27. Sum pixel areas per class in the selected parish; report percentages of assessed eligible vegetation.
28. Export UInt8 GeoTIFF with 255 representing masked/unassessed pixels.